MAST90084 Assignment X

Your Name (Student ID)

2026-04-04

```r
# Reproducibility
library(conflicted)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
```

```r
set.seed(1) # CHANGE TO YOUR STUDENT ID
```

Simple simulation + ggplot example

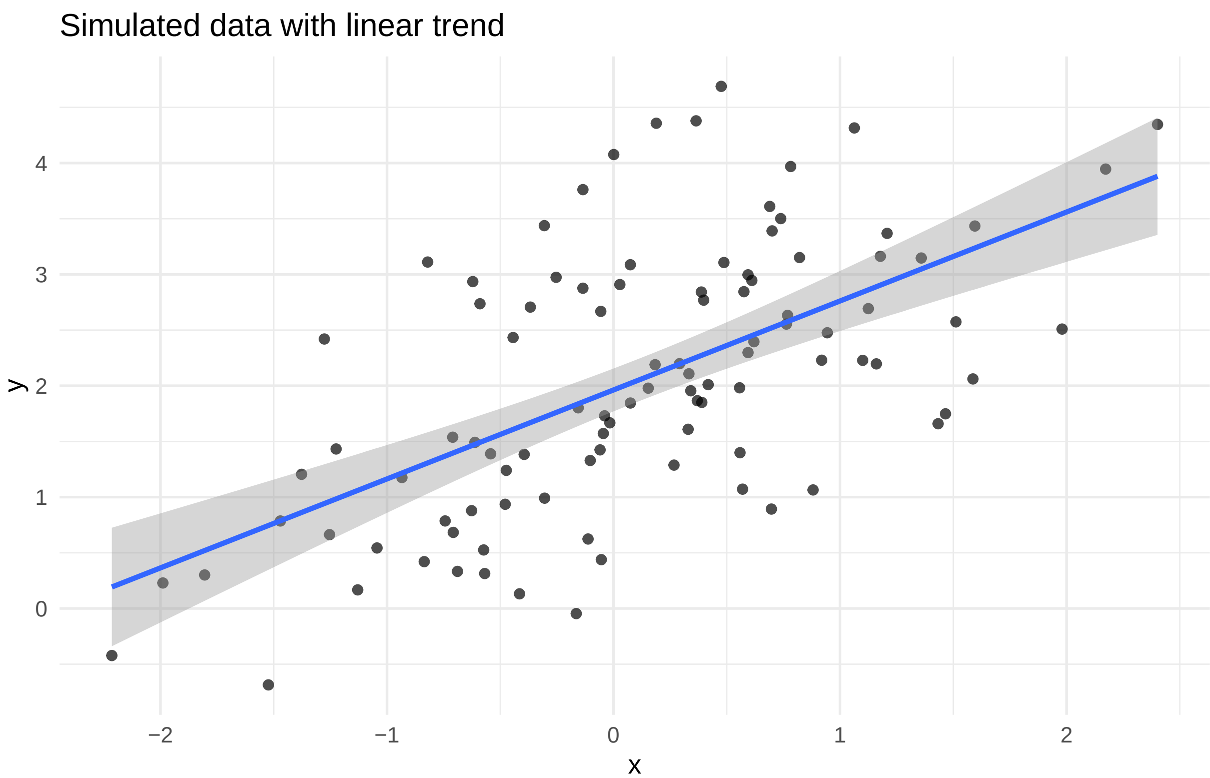
```r
# Simulate basic data
n <- 100
x <- rnorm(n, mean = 0, sd = 1)
y <- 2 + 0.8 * x + rnorm(n, mean = 0, sd = 1)
sim_data <- tibble(x = x, y = y) |>
    arrange(x)

sim_data
```

```
# A tibble: 100 x 2
       x      y
   <dbl>  <dbl>
 1 -2.21 -0.422
 2 -1.99  0.229
 3 -1.80  0.301
 4 -1.52 -0.686
 5 -1.47  0.786
 6 -1.38  1.20
 7 -1.28  2.42
 8 -1.25  0.663
 9 -1.22  1.43
```

```
10 -1.13  0.167
# i 90 more rows
```

```
# Plot with ggplot2
ggplot(sim_data, ggplot2::aes(x = x, y = y)) +
    geom_point(alpha = 0.7) +
    geom_smooth(method = "lm", se = TRUE) +
    labs(
        title = "Simulated data with linear trend",
        x = "x",
        y = "y"
    ) +
    theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



LaTeX examples (quick reference)

Use `equation` for one numbered equation.

$$\hat{\theta} = \arg\max_{\theta} \ \ell(\theta) \tag{1}$$

Use `split` inside `equation` when one equation is too long and needs line breaks, or has a flow of equals signs.

$$
\begin{aligned}
\log L(\beta) = \sum_{i=1}^{n} \left\{ y_i x_i^T \beta - \log\left(1 + e^{x_i^T \beta}\right) \right\} \\
- \frac{\lambda}{2} \beta^T \beta \\
= \sum_{i=1}^{n} \left\{ y_i x_i^T \beta - \log\left(1 + e^{x_i^T \beta}\right) \right\} - \frac{\lambda}{2} \beta^T \beta
\end{aligned}
\tag{2}
$$

Use `gather` for multiple equations that are centered and not aligned at symbols.

$$
\begin{gather}
\mathbb{E}[X] = \mu \tag{3} \\
\mathrm{Var}(X) = \sigma^2 \tag{4}
\end{gather}
$$

Use `align` for multiple equations aligned at `=` (or another symbol via `&`).

$$
\begin{align}
\Pr(Y = 1 \mid x) &= \frac{e^\eta}{1 + e^\eta}, \quad \text{for } \eta = x^T \beta \tag{5} \\
\log\left(\frac{\Pr(Y = 1 \mid x)}{1 - \Pr(Y = 1 \mid x)} \right) &= x^T \beta \tag{6}
\end{align}
$$

You can reference these using `Equation \ref{eq:single}`.

Question 1

(a)

Write your answer here.

```
# Code for Question 1(a)
```

(b)

Write your answer here.

```
# Code for Question 1(b)
```

Question 2

(a)

Write your answer here.

```
# Code for Question 2(a)
```

(b)

Write your answer here.

```
# Code for Question 2(b)
```

Question 3

(a)

Write your answer here.

```
# Code for Question 3(a)
```

(b)

Write your answer here.

```
# Code for Question 3(b)
```

Question 4

(a)

Write your answer here.

```
# Code for Question 4(a)
```

(b)

Write your answer here.

```
# Code for Question 4(b)
```